MENLO SCHOOL WHITTAKER LAB

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Applied Science Research Electric Motor Project Paper

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Abstract

In this project, we created a functional 12-phase 6-pole brushed motor. The motor competes in the categories of running torque, where it achieved 0.29 nm of running torque (3rd); in the unexpected category of efficiency, where it achieved 8.44% efficiency (6th); in the category of rotational speed with a measurement of 1124 rpm. To achieve these metrics, we utilized extremely powerful permanent magnets; spring-loaded graphite brushes, and high uptime commutator panels.

(To be submitted to Dr. Dann, Ph.D.)

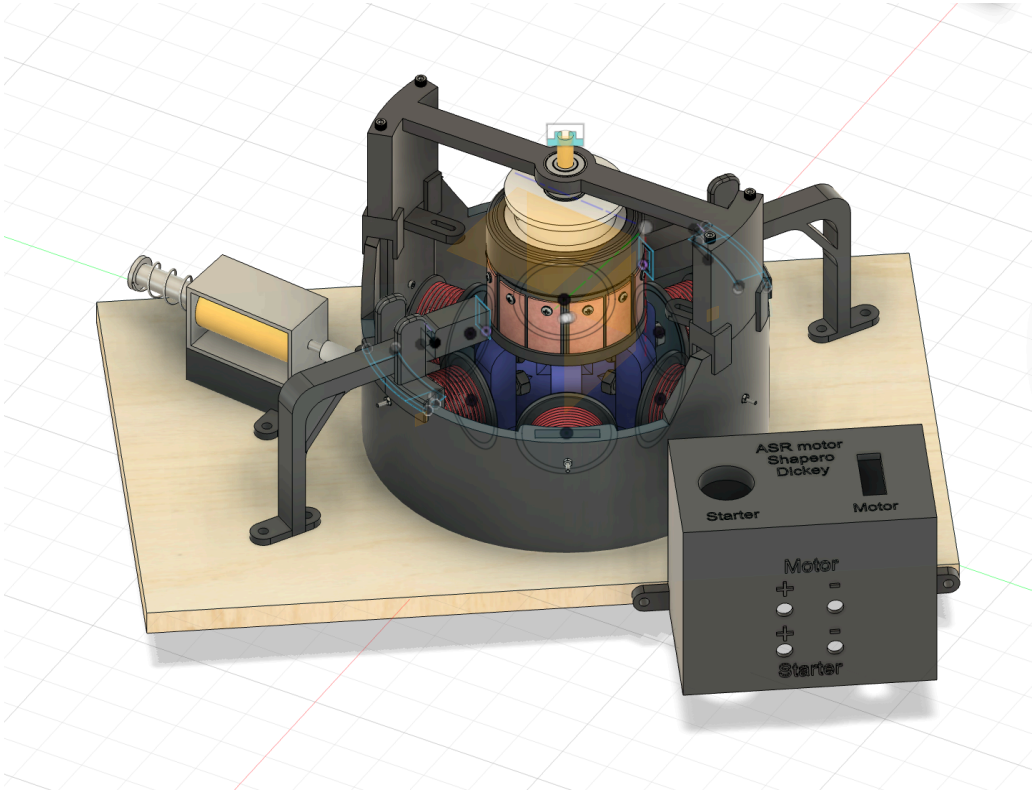
1 Design

Going in, our primary focus and goal was on running torque, and this remained our focus throughout the entirety of the project. However, when making measurements, we discovered that the motor had abnormally high efficiency for a torque motor. The following illustrates design choices made deliberately to increase running torque, which also inadvertently increased motor efficiency.

Initially, we attempted to use ball bearings as our brush, but this proved fruitless, as we were experiencing large amounts of loss. As such, we switched to graphite brushes, which behaved significantly better, reducing loss in electrical arcing and noise. The graphite brushes showed overall significantly lower friction and better conductivity over their metal counterparts.

We also wanted to maximize the number of permanent magnets so that the relative magnetic field strength impacting the electromagnets is as high as possible. This is because closer together permanent magnets (multiple), helps to eliminate “dead zones” at the midpoint between permanent magnets where the magnetic field is so weak that it’s barely helping to push/pull the electromagnets.

The frame for our motor was built and iterated upon throughout the process to continuously stabilize the motor and its components. Initially, there was no full ring frame surrounding the exterior holding the permanent magnets, but there were small individual holders drilled into place, which would shift, move and were not reliably positioned. As such, we created the circular frame to provide significantly more support and reliable positioning of the permanent magnets spaced around the rotor. We later added a top component to help stabilize the rotor itself as it spun, and extra components attaching the brushes directly to the frame to help them from swinging back and forth.



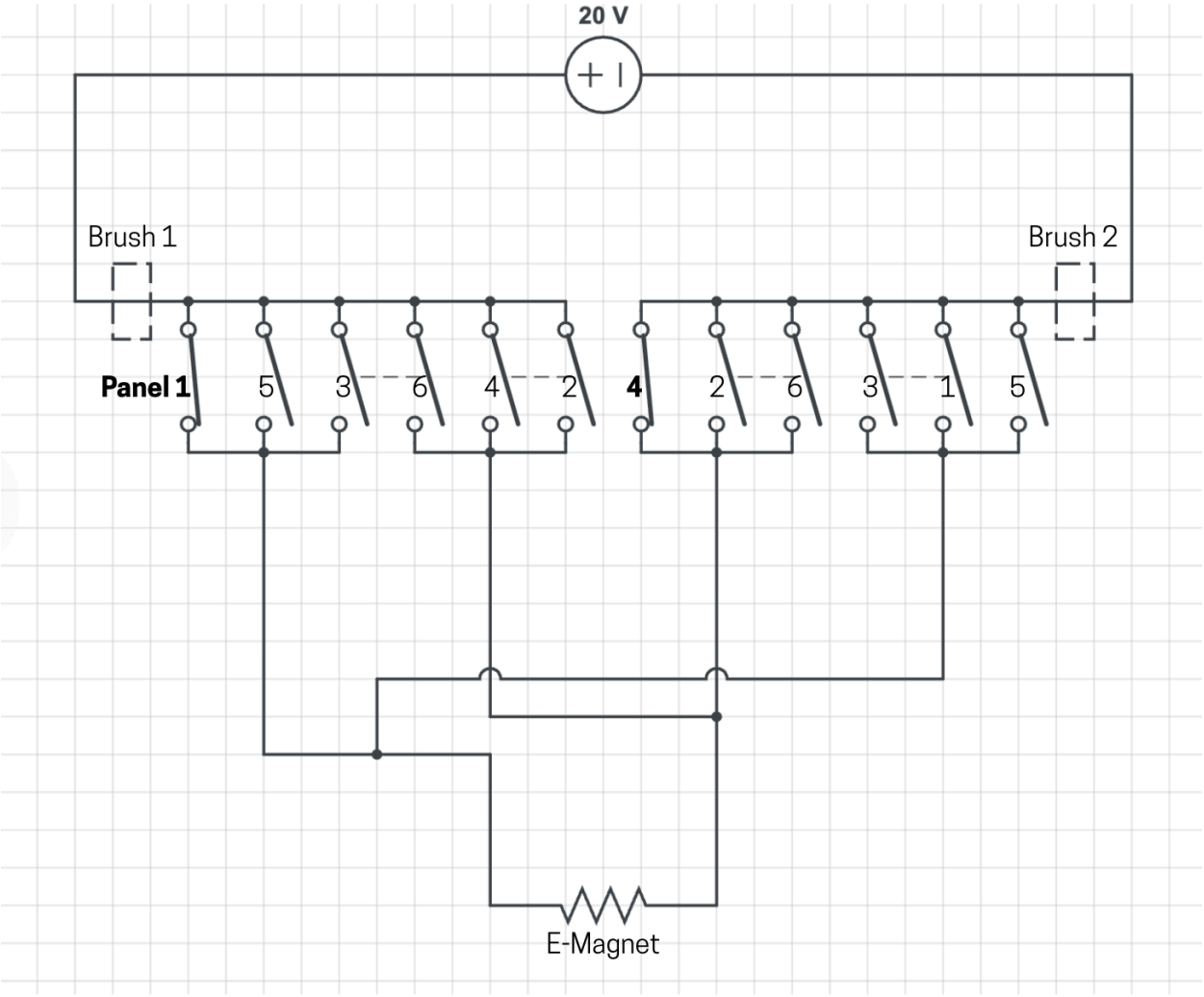


Figure 3a) Above is our circuit diagram following one e-magnet over a full rotation. In the first phase, current flows through panels 1 and 4 as seen above. This is the state at the base position for our motor (see Figure 2a). This state causes each e-mag to repel from the permanent magnets (which is why only one e-mag is shown; they're all doing the same thing; to reduce redundancy). In the second phase, current flows through panel 6 and out through panel 3 (reference Figure 2b). It's important to note here that the current is flowing through the e-mag in the opposite direction so that in this case, it's being attracted to the permanent magnet. In the third phase, current flows through panel 5 and out through panel 2 (reference Figure 2c). The current now switches back to flowing through the e-mag in the same way as the first phase, repelling from the permanent magnets. Each successive phase from hereon is the same as the first 3, but flipped (4->1), (3->6), (2->5).

2 CAD drawings

*All CAD drawings are attached at the end of the paper

Figure 2a)
The drawing presents the overall construction of the motor's various components, including the motor itself, the external solenoid starter, and the control box. The majority of the motor's components are made from 3D-printed PLA or ABS plastic to achieve precision and durability. The motor is secured to a piece of plywood for stability and is connected to a power supply via 2 ports in the control box.

Figure 2b)
The drawing details one of the six solid-state magnets embedded in the case (Figure 2D) of the motor. The magnets are neodymium N52 and have a max holding force of up to 136lbs per magnet when attached and 38lbs of holding force when attached to a steel plate. These specific magnets were chosen due to their incredible force, which is essential for maximizing torque.

Figure 2c)
This drawing details one of the 6 electromagnets within the motor. The electromagnets were constructed using a ¼ in threaded steel rod with 2 acrylic disks with holes cut attached at a distance of 1in apart (secured with nuts). Then, 4oz of 24-gauge copper wire was wrapped around the rod as tightly as possible to achieve maximum force. Afterwards, the copper coil was wrapped in electrical tape, the front panel was taken off and replaced with a thinner plastic panel, and the excess threaded rod was removed. Washers were added as spacers between the rotor and the electromagnet so the distance could be adjusted easily.

Figure 2d)
This drawing details the casing of the motor. The case is printed out of PLA-CF plastic and has to be glued together after the print failed halfway through. The case has places to slot in the magnets as well as attachment points for brushed (however, they did not end up being used). The design of the case is optimized for maximum strength, yet does not use more material than necessary

Figure 2e)
The drawing showcases the commutator of the motor, one of the most essential parts. The overall frame of the commutator is 3D printed out of PLA-CF plastic for optimal weight and strength. There are 12 panels of sheet copper around the commutator, cut into identical rectangles. The panels were bent uniformly using a 3D printed vise jig. After securing the panels to the commutator with screws (which also act as the conductive wiring), the entire commutator was coated in epoxy resin. The resin was later sanded down flat so that the entire surface of the

commutator would be smooth and not have any bumps that could affect the motor's brushes. The commutator is also press fit and secured with screws onto both the spool stabiliser and the electromagnet holder. It is also important to note that all electronics in the motor are connected via a modified protoboard housed within the commutator.

Figure 2f)
This drawing details the spool stabiliser. The component has 2 jobs, one of which is to act as the measurement tool for measuring running torque and efficiency, and the other is to stabilize and protect the shaft from components such as the electronics in the commutator.

Figure 2g)
The graphite brush holder is an essential component of the motor because it delivers current to the inner rotor and electromagnets. The holder is secured to the wood outside the casing and then stabilized against the casing with an additional component. The actual graphite brush was bought from Amazon and is slid into the holder along with a spring so that the brush will always have pressure against the commutator. The brush holder is printed out of ABS, as there were issues regarding melting when materials such as PLA were used.

Figure 2h)
The electromagnet holder secures the 6 e-mags to the rotor. The holder is 3d printed from ABS for strength. The magnets are secured using nuts, and the entire component is attached to the shaft with 2 press-fit bearings on top and bottom.

Figure 2i)
The 3 electromagnet holder was a prototype holder for version 2 of the motor. Version 2 only had 3 electromagnets, hence the need for only 3 slots for emags. The design of the holder allows for it to be swapped into the final motor at any point if desired.

Figure 2j)
The drawing presents the overall design of the version 2 motor. This motor was designed to be solely for testing purposes, which is why nothing is connected permanently. The motor was used to test different distances and electromagnets and was essential in finding the right layout for the final design.

3 Results

Running Torque

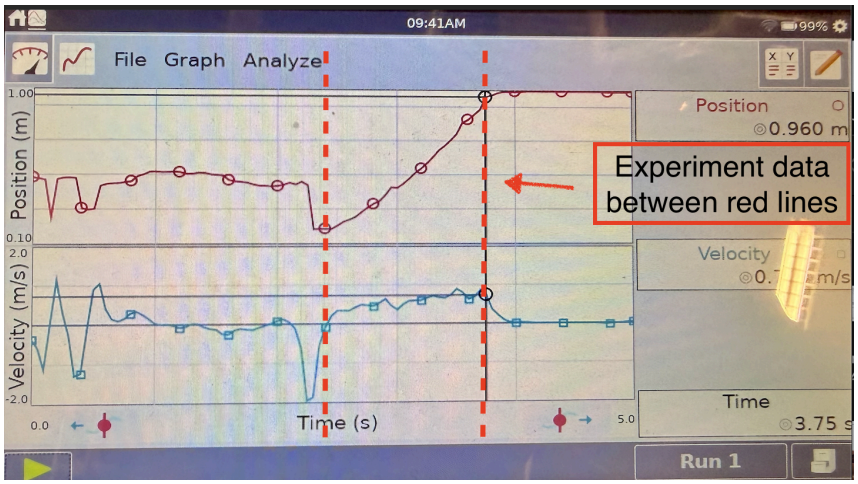


Figure 4) Acceleration vs time of a weight being lifted by the motor. The top graph shows the position of the weight relative to the device measuring which is positioned underneath the weight. As the motor runs, the weight is visibly lifted away from the ground at a constant acceleration from 0[1]

Time to lift weight: 1.02seconds
Distance lifted = Δx = .85m
Mass = 1.00kg
Spool Radius = 1in = .0254m
Acceleration of gravity = 9.8m/s^2

Equation for acceleration -> $\Delta x = 1/2at^2$
$.85 = 1/2a(1.02)^2$
$.85 = 1/2a(1.04)$
$.85/(1.04*.5) = a$
$a = 1.634 m/s^2$

Equation for Force -> $F_T - mg = ma$
$F_T = mg + ma$
$F_T = (1kg)(9.8m/s^2)+(1kg)(1.634m/s^2)$
$F_T = 11.434n$

[1] The bottom blue graph showcases that at the advent of the weight rising, velocity of the weight was 0

Equation for Force around circle -> torque(t) = F_T * r
t = (11.434n)(0.0254m)
t = 0.2904nm

Our method for measuring the running torque of our motor included attaching a 2 inch diameter spool to the top of the motor, with string running over the top of a pulley situated at the edge of the plyboard holding the motor and its components. This string was attached at the bottom to a known weight sitting ~1 meter below the pulley at rest. The motor was allowed to run with a minimal amount of tension holding the string. An ample amount of slack string was provided to allow the motor to pull the weight with its running torque and not from a zero point. The time that it took for the motor to raise the weight a meter from its rest was then recorded.

Rotational Speed

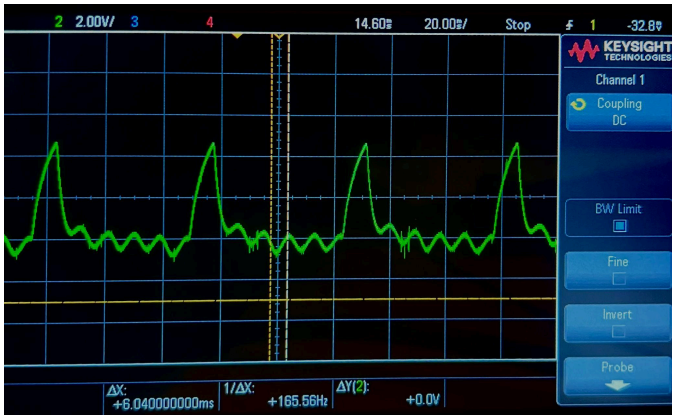


Figure 5) Oscilloscope reading for rotational speed. This reading shows 2 states: high when the paper attached to the commutator was being read, and low when it was not. Due to this, the time between each peak is the cycle of the motor.

Amount of trigger points in one revolution = 1
Time interval between peaks = 53.4ms

RPM equation -> 60sec / # of sec for 1 revolution = RPM
53.4ms = 0.0534sec
RPM = 60sec/0.0534sec
Rotational speed = 1124RPM

To measure the rotational speed, we attached a small flap of tape to the top part of our commutator so that it would protrude slightly. We then set up the laser attachment for the oscilloscope, positioning it so that the laser will hit the tape as it passes by. By running both the motor and oscilloscope in tandem, we were able to see the time it took for one cycle of the tape (and therefore one rotation, because it's essentially either on/off) to pass. By dividing 60 by the cycle time, we were able to calculate the rotations per minute (RPM) of the motor.

Efficiency

Measuring the power consumption of the motor

Highest recorded voltage = 21.5v
Highest recorded current = 5.25amps

Equation for power P = V * I
P = 21.5v * 5.25a
P = 112.9watts

Measuring the power required to lift the weight

Time to lift weight: 1.02seconds
Distance lifted = Δx = .85m
Mass = 1.00kg
Acceleration of gravity = 9.8m/sˆ2
Acceleration of Object =

Equation for energy -> E = mgh
E = (1.00kg)(9.8m/sˆ2 + 1.634m/sˆ2[2])(.85m)
E = 9.72J
P = J / t
P = 9.72J / 1.02sec
P = 9.53w

Equation for Efficiency -> P_{output} / P_{input} *100% = Efficiency
Efficiency = 9.53w/112.9w * 100%
Efficiency = 8.44%

[2] This value was calculated when finding the running torque.

The process to measure efficiency involved recording the highest recorded voltage and current when lifting a known weight a specific height over time. The voltage and current were then multiplied by one another to get the power (Pin = VI). With the known amount of power needed to lift the weight (Pout = mgh/t), we could then determine our efficiency percentage by dividing the lifting work by the electrical input power and multiplying by 100.

Table 1: List of motor's final calculated specifications

Category	Result
Rotational Speed	1124rpm
Running Torque	.29nm
Startup Torque	N/a[3]
Efficiency	8.44%

4 Conclusion

All things considered, our motor was a large success and works very well, earning multiple records on the Menlo School Hall of Fame leaderboard. In terms of next steps, we would try to find ways to reduce some of the unnecessary weight encumbering the rotor, and find ways to further reduce tolerances between electromagnets and solid magnets.

[3] The motor self started using a solenoid making an attempt at running torque unfair. The motor would also catch on fire whenever on but not actively spinning.

5 Bibliography

J. Owen, “How Does an Electric Motor Work? (DC Motor),” YouTube, video, Year. [Online]. Available: https://www.youtube.com/watch?v=CWulQ1ZSE3c

Perplexity AI, “DIY Brushed and Brushless Motor Design Q&A Thread (User: torque-focused design)”, Private technical conversation, Sep.–Oct. 2025. [Online]. Available: https://www.perplexity.ai/chat/ (accessed Oct. 16, 2025).

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